

Executive Summary

India's law enforcement is investing in **dark-web intelligence** to counter cybercrime. In 2023 the Ministry of Home Affairs (MHA) and partners launched *KAVACH-2023*, a national AI hackathon for cybercrime solutions [8†L25-L33]. Among the winners was **Team Cyph3r** (KJ Somaiya engineering), whose AI-powered "Dark Web Crawler" (codenamed *Cyph3r*) extracts actionable data from deep and dark-web sources [16†L38-L44] [46†L68-L75]. MHA's Indian Cyber Crime Coordination Centre (I4C) confirms it is **deploying AI-based tools** to monitor dark-web marketplaces, scam sites and fraud networks [33†L93-L100]. By contrast, we found **no public information** on "CID Track" – presumably an internal criminal-intel tracking system – suggesting it may be classified or not yet disclosed.

KAVACH/Cyph3r and related platforms aim to collect data (Tor forums, markets, paste sites, encrypted chats, cryptocurrency trails) and apply AI/ML for analysis. They resemble global dark-web intel tools (e.g. DarkOwl, SecNinjaz WebMine) that **continuously crawl hidden services, correlate identities, score risks and generate investigation leads** [73†L132-L145] [75†L148-L156]. Our review covers their origins, architecture, capabilities/limits, use-cases, and legal/ethical issues. Key findings: (1) *Cyph3r* is a proof-of-concept crawler from a government hackathon, not a fielded nationwide system [16†L38-L44] [46†L68-L75]. (2) Officially, I4C and agencies use *proprietary AI systems* (e.g. CDAC's child-abuse monitor) for dark-web intelligence [33†L93-L100], but details are scant. (3) No open code or patents were found for these tools. (4) Capabilities include large-scale Tor crawling and data extraction (AI/ML classifiers, OCR, entity linkage) but face limits (encryption, attribution, jurisdictional law differences). (5) Robust oversight is needed to balance crime-fighting with privacy and legal safeguards.

Recommendations: Law enforcement should validate any dark-web intel system through pilot deployments and judicial review, ensure compliance with Indian cyber laws (IT Act, criminal procedure) and data-protection norms, and promote transparency (e.g. publish technical standards or audit reports). Cross-agency data-sharing and training (especially at state CID units) are also critical.

Background: Dark-Web Intelligence and Law Enforcement

The **dark web** comprises Tor/I2P hidden services inaccessible via normal search engines. It hosts illicit marketplaces and forums for drugs, weapons, fraud and child exploitation. Over 2.5 million users reportedly connect daily to Tor, with ~7% accessing hidden services [73†L104-L112]. Criminal networks (e.g. India's recent "Team Kalki" drug case) have used darknet forums plus encrypted messaging to run pan-India smuggling rings [73†L64-L72]. Law enforcement worldwide counters this by using specialized tools that crawl and index onion sites, extract entities (users, products, transactions), correlate across sources, and flag high-risk content [73†L132-L145].

Indian agencies have begun adopting such methods. MHA told Parliament that its I4C centre "*uses AI-based tools to monitor the dark web, scam websites and fraud networks*" for intelligence [33†L93-L100]. Similarly, a CDAC-developed **Proactive Monitoring Tool** (PMT) uses AI to scan the web for child sexual exploitation content [33†L85-L93]. Kerala Police's Cyberdome and other cyber cells also conduct *dark-web monitoring* and investigations. However, details on specific platforms (especially for general law enforcement use) are sparse in public sources.

Figure: SecNinjaz (cybersecurity vendor) depicts a typical dark-web intelligence workflow: crawling onion sites, normalizing data, correlating entities, scoring leads, and reporting to investigators [73†L132-L145] . Such architecture underpins solutions like “Cyph3r” and commercial services. [76†embed_image] *Figure: Conceptual dark-web intelligence platform architecture (source: SecNinjaz) [73†L132-L145] .*

KAVACH-2023 Hackathon and Team Cyph3r

In February 2023 the Indian government launched **KAVACH-2023**, a national hackathon on cybersecurity organized by MHA (Bureau of Police R&D and I4C) in collaboration with AICTE and the Education Ministry [8†L25-L33] . Over 100 teams competed on challenges ranging from fake news and women’s safety to **Dark Web intelligence** [8†L47-L55] [16†L38-L44] . One problem statement (KVH-06) tasked competitors to develop a “*Dark Web Crawler*” providing actionable data to police.

Team Cyph3r from K J Somaiya School of Engineering won this category. Their solution is an **AI-enabled darknet crawler** that collects data from hidden web forums and marketplaces and extracts leads for investigators [16†L38-L44] [46†L68-L75] . Descriptions indicate it was built during the 36-hour finale in Bhopal: an autonomous scraper/parser that identifies relevant content (e.g. illicit trading, mule accounts) via machine learning. The team’s mentors were Dr. Irfan Siddavatam and Prof. Ashwini Dalvi [16†L38-L44] [46†L68-L75] . The government highlighted their “*AI-powered Dark Web Crawler*” as a notable hackathon outcome [46†L68-L75] .

Who: Team Cyph3r comprised six students (Siddhika Rai et al.) and was sponsored by academia. The hackathon itself was organized by MoH (I4C/BPR&D) and AICTE [8†L25-L33] [16†L38-L44] . No private contractor was mentioned in sources.

Where/When: Developed August 2023 at the IES Bhopal nodal center (KAVACH-2023 grand finale) [16†L38-L44] [46†L68-L75] . Deployment appears limited to the hackathon demonstration; we found no evidence it entered active police systems. (It’s possible follow-up R&D occurred privately, but none was publicized.)

Why: The tool addresses the use-case of mining the deep/dark web for crime intelligence – a mandate under I4C’s coordination of cybercrime fighting. A parliamentary report notes I4C’s focus on AI-driven monitoring of fraud and scam networks [33†L93-L100] . Thus *Cyph3r* was created to meet exactly these law-enforcement needs: extracting “**actionable insights**” from hidden web data to aid investigators [16†L38-L44] [46†L68-L75] .

CID Track (Unknown)

We found *no authoritative information* on a system called “**CID Track**”. No government press release, tech report or patent by that name appears in searches. It may be:

- An internal tool/operation (perhaps of a state CID or central agency).
- A project codename not disclosed publicly.
- A misunderstanding of another system.

Assumption: If “CID Track” refers to **Criminal Intelligence Department tracking**, one might speculate it is a database or monitoring platform used by police to link cyber leads with criminal records. Without

evidence, we can only note its absence. (By contrast, “Cyph3r” is documented via the hackathon.) We encourage law enforcement agencies to clarify such tool-names or publish documentation if possible.

Architecture and Technologies

Tools like Cyph3r and any CID surveillance platform would rely on multi-layer architectures similar to global dark-web intel systems [73†L132-L145]. Key components include:

- **Collection:** Automated crawling of onion links (Tor/I2P), deep-web forums, paste sites, and even public social media. For example, *DarkOwl* claims a database of ~18 billion darknet emails and continuous crawling at ~227,500 pages/hour [75†L132-L142]. *Cyph3r* presumably gathers forum posts, marketplace listings, chat channels etc.
- **Processing/Normalization:** Extracting entities (names, addresses, cryptocurrencies, images) from unstructured text. OCR may be used for images. Data is cleaned and standardized.
- **Correlation & Scoring:** Matching suspects or groups across sources, linking alias usernames, assigning confidence scores. SecNinjaz notes this is what separates raw data from “prioritized, connected leads” [73†L132-L145]. Machine learning models (NLU/NER) can classify content (e.g. identifying mule banks or trafficking keywords).
- **Analytics & Reporting:** Generating dashboards or case files for investigators. This could integrate GIS mapping of crime routes or timeline graphs. SecNinjaz emphasizes that final reports must be legally credible (not just screenshots) [73†L130-L145]. The hackathon solution likely outputs alerts or summaries of suspicious content.
- **Data Sources:** Besides Tor/I2P, systems tap underground channels (Telegram invite links, encrypted forums), anonymous crypto-tumblers, and even leaked Dark Web data dumps. Some use *open-source intelligence* from the clear web (e.g. social media footholds) to seed dark-web searches.

Technologies: Python/Node crawlers, Tor/I2P proxies, AI/ML frameworks (TensorFlow, PyTorch) for language and image analysis, graph databases for linking entities, and possibly Big Data platforms (Elasticsearch, Hadoop) for indexing. The hackathon team may have used open-source tools (e.g. Tor controllers, web scraping libs) plus custom ML models. All such tools require secure deployment (likely on isolated networks) and possibly *Mirai* style virtual machines.

Capabilities and Limitations

- **Capabilities:** *Cyph3r* and similar platforms can monitor hidden services continuously, flag stolen credentials or contraband postings, and produce suspect profiles [16†L38-L44] [33†L93-L100]. SecNinjaz notes that only automated systems can scale to hundreds of thousands of hidden-service users [73†L104-L112]. Features may include keyword alerts (e.g. brandishing national threats), link analysis (e.g. same vendor on multiple markets), and even facial-recognition on child abuse images (for related initiatives). In India’s context, they help trace mule-money networks and trace phishing rings operating via dark-web channels.

- **Limitations:** *Anonymous Tor design:* True anonymity networks hinder attribution. A user's identity remains hidden unless an operator leaks info. Encryption and ephemeral addresses mean content can vanish. Manual analysis often remains necessary. False positives are common (e.g. decoy listings). Moreover, *Cyph3r* was only a prototype; without integration into police IT systems, its outputs are unused. Scalability and maintenance are also concerns: crawling large portions of the dark web requires significant bandwidth and compute.

Legally, any infiltration or undercover chat (if attempted) requires caution; simply scraping public onion pages is usually allowed, but active login or hacking would cross legal lines. India's **IT Act** and criminal procedure require authorized warrants for intrusive surveillance, though gathering public dark-web postings might be treated like open-source intel. Still, agencies must ensure compliance with privacy laws (e.g. not over-reaching on personal data) and hand over evidence properly chain-of-custody.

Deployments, Cases and Outcomes

No official "case studies" were released for *KAVACH/Cyph3r*. It has not been reported as field-deployed by police or central agencies. The hackathon winner was praised but ultimately it appears to remain a proof-of-concept. We found no news of it solving an actual investigation.

Separately, Indian agencies have had **dark-web successes**. For example, in Mar 2026 the Narcotics Control Bureau credited dismantling "Team Kalki", a darknet-based LSD/MDMA trafficking network, by tracking its forum vendor reputation and encrypted chat channels [73†L64-L72] . This shows police using dark-web intel (though details of the tools used were not publicized).

Internationally, similar tools (like *DarkOwl* or *Group-IB* platforms) have assisted in busting ransomware and child-abuse rings. In the U.S., agencies use blockchain analysis combined with darknet data to freeze illicit funds. Though Indian specifics are scarce, it's clear that agencies aim for proactive monitoring: e.g., I4C's statements emphasize *faster, AI-enabled decision-making* in cybercrime [33†L59-L68] .

Documentation, Code Repos and Leaks

No official manuals or code for *Cyph3r* or any "CID Track" were found. Searches of Indian patent databases and open code repositories (GitHub, GitLab) yielded nothing. *Team Cyph3r* may have an internal codebase, but nothing was published. We found no media leaks, Wikileaks cables, or dark-web chatter referencing these tool names. The only documentation available is the hackathon press kit [8†L25-L33] and news coverage [16†L38-L44] [46†L68-L75] .

By contrast, vendor tools (DarkOwl, etc.) have whitepapers and data sheets (e.g. DarkOwl's website cites their dataset sizes [75†L132-L142]). For policy and oversight, India has issued no public standards for dark-web surveillance tools yet. This gap suggests a need for transparency or at least independent audits of any such system used by police.

Legal and Ethical Issues

Dark-web intelligence sits at a legal and ethical gray zone. Key issues include:

- **Privacy:** Users on Tor assume anonymity. Even if users are criminals, automated scraping could inadvertently collect innocuous personal data (e.g. bystanders' info or grey-market buyers). India's evolving privacy law (the Digital Personal Data Protection Act, 2023) emphasizes consent and minimal data usage; broad mass surveillance may conflict with these norms.
- **Due Process:** Any evidence gathered must be admissible. Indian law (IT Act 2000, CrPC 1973) requires warrants for interception of communications. Pure OSINT from websites is generally lawful, but if an investigator uses malware or deception to infiltrate private sections, it risks illegality. Chains-of-custody and verification (verifiable hashes, screenshots) are crucial to ensure court acceptance.
- **Oversight:** Without clear statutes, there is risk of mission creep. A tool like *Cyph3r* should be governed by clear authorizations (e.g. sanction by Cyber Appellate Tribunal if encryption is involved). Building "big data" profiles could raise concerns similar to social media scraping (discrimination, bulk data retention). The government has yet to specify oversight mechanisms for AI-driven surveillance, so robust internal rules are advised.

Internationally, experts warn that dark-web policing can be a "neighborhood watch" but also requires checks [33†L59-L68] . We recommend that usage of KAVACH/Cyph3r or CIDTrack be limited to declared investigation scopes (e.g. terrorism, child abuse) and that legal review boards evaluate any new intel system.

Comparative Table of Dark-Web Intelligence Tools

Tool / Platform	Features / Capabilities	Data Sources	Legal Constraints
KAVACH "Cyph3r" (prototype)	AI-based darknet crawler extracting actionable leads (keywords, profiles) [16†L38-L44] . Likely uses ML for classification; no public UI.	Tor/I2P hidden services (forums, markets), deep-web pages. Potentially Telegram/IRC links. (<i>Hackathon project; scope unknown.</i>)	Not formalized. Any use by police must comply with IT Act surveillance rules. Likely limited to open-access posts (not intercepting encrypted comms).
"CID Track" (undisclosed)	Unknown. Presumably a criminal-intel tracking system, possibly linking cyber leads with police records. May automate case linking.	If it exists, possibly law-enforcement databases, open web, social media, phone records, financial transactions.	Unknown; presumably internal to police. Would be bound by internal rules; could involve sensitive personal data.

Tool / Platform	Features / Capabilities	Data Sources	Legal Constraints
Commercial Platforms (e.g. DarkOwl Vision) [75†L148-L156]	Continuous darknet monitoring, search APIs, entity extraction, risk scoring. Features include breach alerts, malware & ID fraud detection. AI-assisted analysis to reduce manual work.	Billions of collected records: forums, marketplaces, leaks. Includes indexed Tor pages, pastebin sites, deep-web caches [75†L132-L142] .	Operates on publicly leaked data; generally legal for businesses. Law enforcement use is by contract. Privacy rules depend on data sources; often treated as OSINT.
Vendor Intelligence (e.g. SecNinjaz WebMine) [73†L132-L145]	Multi-layered platform with continuous crawler, NLP & graph analysis. Authorized user workflow, deliver case files for court. Emphasizes reliable scoring of leads.	Onion sites (forums, markets, paste), plus possibly intelligence from partners. Integrates open web & CSEAM data.	By design, focuses on correlating public hidden-service data. Ensures audit logs. Must adhere to judicial evidence standards (SecNinjaz claims “court-acceptable” reports [73†L132-L145]).
Open-Source Tools (e.g. Maltego, TorBot)	Manual or semi-automated OSINT gathering. Transform-based queries linking known alias/domains. No turnkey analysis.	Mix of public web, Tor (if configured), known data breaches, social profiles.	Usage is fully legal for open data, but analysts bear responsibility for not hacking private systems.
Custom In-House Systems (e.g. CDAC PMT, “Surakshini” Dashboard)	Focused tools (e.g. PMT for CSEAM detection [33†L85-L93]). Dashboard collating complaints and takedown status.	PMT crawls known risky sites, child victim keywords. Surakshini uses hash-sharing for CSEAM.	Operated by certified investigators. Falls under IT Act provisions; uses only reported/offered content.
Blockchain Analytics (CryptoTracers)	Not dark-web per se, but often integrated: tracks cryptocurrency transactions (useful for darknet payments).	Public blockchain ledgers (Bitcoin, Monero mixing data). Combined with known marketplace wallets.	Usually legal to analyze since blockchains are public records. Care needed when linking to real identities (may need warrants).

Table: Comparison of selected dark-web intelligence tools. “Data Sources” indicates the types of inputs each uses; “Legal Constraints” notes typical limitations under Indian law. Sources: Official hackathon and news for KAVACH/Cyph3r [16†L38-L44] [33†L93-L100] ; vendor sites and analyses for others [73†L132-L145] [75†L148-L156] .

Timeline

- **Feb 2023:** MHA launches KAVACH-2023 hackathon (MoE/AICTE/BPR&D/I4C) 【8†L25-L33】 .
- **Aug 2023:** Team Cyph3r wins KAVACH dark-web challenge; “Dark Web Crawler” announced 【16†L38-L44】 【46†L68-L75】 .
- **Mar 2026:** Parliamentary report notes I4C’s use of AI for dark-web monitoring 【33†L93-L100】 .
- **Mar 2026:** NCB announces bust of “Team Kalki” darknet drug ring (SecNinjaz analysis) 【73†L64-L72】 .

Entity-Relationship Diagram

Below is a high-level schematic of stakeholders, data flows and integrations in India’s dark-web intelligence ecosystem:

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graph LR
    subgraph Agencies
        MHA[MHA / BPR&D / I4C]
        Police[Law Enforcement Agencies]
        CDAC[CDAC / Cyberdome]
    end
    subgraph Academic
        KAVACH[KAVACH Hackathon]
        Cyph3r[Team Cyph3r]
    end
    subgraph DataSources
        Darkweb[Dark Web (Tor/I2P Forums, Markets)]
        Social[Social Media / Open Web]
        Financial[Crypto/Banking Data]
    end
    subgraph Platforms
        AItool[AI/ML Analysis Engine]
        PMT[Proactive Monitoring Tool]
        Surakshini[OCWC Dashboard]
        Vendors[Vendor Tools (DarkOwl, etc.)]
    end

    MHA --> KAVACH
    KAVACH --> Cyph3r
    Cyph3r -->|feeds| AItool
    Darkweb --> AItool
    Social --> AItool
    Financial --> AItool
    CDAC --> PMT
    PMT --> AItool
    AItool --> Police
    Police --> AItool
    Vendors --> AItool
```

MHA --> Surakshini
Surakshini --> Police

Diagram (Mermaid) – Entities and data flows in India's dark-web intel framework. The hackathon KAVACH produced Team Cyph3r's crawler; I4C/MHA and vendors provide AI tools; CDAC's PMT and OCWC "Surakshini" supply specialized data; police agencies consume the outputs.

Methodology

This report synthesizes open-source intelligence from official releases, news reports, academic sources, and vendor literature. We searched government press portals (PIB, MHA reports), educational and news sites for "KAVACH" and "Cyph3r" mentions; reviewed KAVACH hackathon documents [8†L25-L33] and media coverage [16†L38-L44] [46†L68-L75]; examined the Parliamentary committee report via press articles [33†L59-L68] [33†L93-L100]; and surveyed cybersecurity vendor analyses for context [73†L132-L145] [75†L148-L156]. Where specific information was unavailable (e.g. "CID Track"), we noted the lack of sources. All cited statements are supported by the connected references listed.

Recommendations

- **Transparency:** Publish at least schematic descriptions (not necessarily code) of any law-enforcement dark-web tools, to allow independent scrutiny.
- **Legal Review:** Ensure compliance with IT Act and judiciary oversight. Mandate warrants or approvals for any active intrusions.
- **Privacy Safeguards:** Anonymize or delete data on uninvolved persons; use data minimization.
- **Inter-Agency Sharing:** Connect dark-web intel (Cyph3r, etc.) with existing crime databases (FIRs, banking alerts) via secured networks.
- **Capacity-Building:** Train CID and cybercrime officers on dark-web analysis, and allocate budgets for proven platforms (possibly procure vendor solutions).
- **Research:** Fund follow-ups on hackathon prototypes to mature them (e.g. convert *Cyph3r* from prototype to pilot program).
- **Public Policy:** Consider enacting guidelines on AI surveillance and designate a government body for oversight of cyber-intel tools.

Sources: Authoritative press releases, academic coverage, and industry reports have been cited throughout. For example, KAVACH hackathon details are from PIB [8†L25-L33] and media accounts [16†L38-L44] [46†L68-L75], while official I4C intentions are from a New Indian Express report quoting MHA [33†L93-L100]. Vendor analyses (SecNinjaz, DarkOwl) provide technical context [73†L132-L145] [75†L148-L156]. Any gaps (notably on "CID Track") are noted as such. All direct quotes and data are properly cited.